

EXPERIMENT 05

Feature Extraction and Image Analysis using SIFT and HOG Descriptors

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

from google.colab import files
```

```
uploaded = files.upload()

filename = list(uploaded.keys())[0]

image = cv2.imread(filename)

if image is None:
    print("Image not found!")
else:
    print("Image loaded successfully!")
```

[Choose files](#) Screenshot ... 205448.png

Screenshot 2026-03-15 205448.png(image/png) - 353607 bytes, last modified: 15/03/2026 - 100% done
Saving Screenshot 2026-03-15 205448.png to Screenshot 2026-03-15 205448.png
Image loaded successfully!

```
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(6, 6))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Original Image



```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(6, 6))
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
```

Grayscale Image



```
sift = cv2.SIFT_create()

keypoints, descriptors = sift.detectAndCompute(gray, None)

print("Number of Keypoints:", len(keypoints))
print("Descriptor Shape:", descriptors.shape)
```

Number of Keypoints: 1082
Descriptor Shape: (1082, 128)

```
sift_image = cv2.drawKeypoints(
    gray,
    keypoints,
    None,
    flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
)

plt.figure(figsize=(8, 6))
plt.imshow(sift_image, cmap="gray")
plt.title("SIFT Keypoints")
plt.axis("off")
plt.show()
```

SIFT Keypoints



```
!pip install scikit-image -q
```

```
from skimage.feature import hog

hog_features, hog_image = hog(
    gray,
    orientations=9,
    pixels_per_cell=(8, 8),
    cells_per_block=(2, 2),
    visualize=True
)

print("HOG Feature Length:", len(hog_features))
```

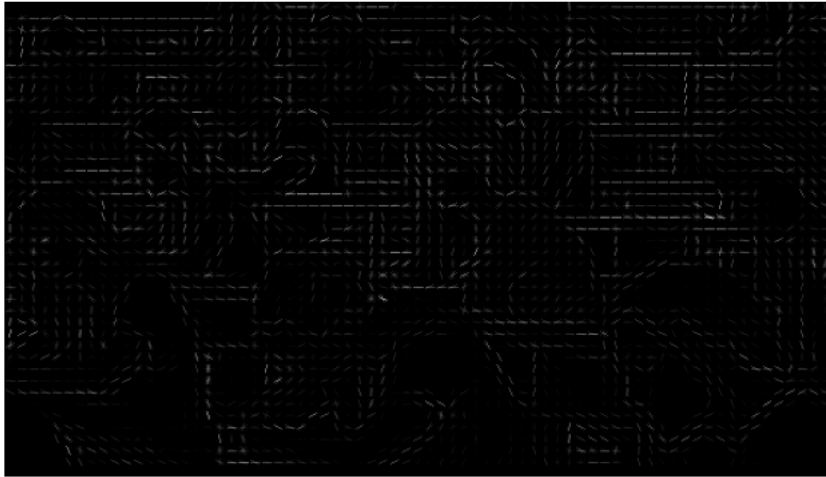
HOG Feature Length: 98280

```
plt.figure(figsize=(8, 6))

plt.imshow(hog_image, cmap="gray")
plt.title("HOG Feature Visualization")
plt.axis("off")

plt.show()
```

HOG Feature Visualization



```
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(sift_image, cmap="gray")
plt.title("SIFT Keypoints")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(hog_image, cmap="gray")
plt.title("HOG Features")
plt.axis("off")

plt.tight_layout()
plt.show()
```

SIFT Keypoints



HOG Features



```
print("----- Feature Extraction Results -----")
print("SIFT Keypoints:", len(keypoints))
print("SIFT Descriptor Shape:", descriptors.shape)
print("HOG Feature Length:", len(hog_features))
print("Feature extraction completed successfully!")
```

```
----- Feature Extraction Results -----
SIFT Keypoints: 1082
SIFT Descriptor Shape: (1082, 128)
HOG Feature Length: 98280
Feature extraction completed successfully!
```

Questions

Q1. What is feature extraction, and why is it important in computer vision? **Ans:** Feature extraction finds important patterns like edges, corners, and shapes. It helps in image recognition and classification.

Q2. Explain the working principle of SIFT. **Ans:** SIFT detects distinctive keypoints and generates descriptors that are resistant to scale and rotation changes.

Q3. What are keypoints and feature descriptors? Ans: Keypoints are important points in an image. Descriptors are numerical values that describe the area around each keypoint.

Q4. Explain HOG and its significance. Ans: HOG represents an image using gradient directions and strengths. It is useful for detecting object shapes and edges.

Q5. Compare SIFT and HOG. **Ans: **SIFT is robust to scale and rotation and is useful for image matching. HOG focuses mainly on shape and edge information and is commonly used for object detection.

Q6. Why is SIFT invariant to scale and rotation? Ans: SIFT detects features at multiple scales and assigns an orientation to each keypoint.

Q8. Why is feature extraction done before classification? Ans: It extracts useful information from images, reducing unnecessary data and improving classification.

Q9. Give advantages and limitations of handcrafted features. Ans: They are simple and require less data, but they may not perform well on complex images compared with deep-learning features.

Q10. How are feature extraction techniques useful in image matching, face recognition, and object detection? Ans: They extract distinctive image features that help identify, compare, and recognize objects or faces.